Surrogate-Conditioned Benchmark Fragility in Block-Scale Urban Energy Design: A Comparative Study of Policy Learning and NSGA-II

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Abstract

Surrogate-assisted optimization is increasingly used in block-scale urban energy design, yet benchmark conclusions can become fragile when optimizer ranking depends on the selected surrogate artifact rather than on direct physical evaluation. This study examines that risk by comparing a Deep Deterministic Policy Gradient (DDPG) workflow and fair-budget NSGA-II in a shared surrogate environment for urban morphology design using three energy-related indicators: Energy Use Intensity (EUIt), Energy Generation (EG), and Sunlight Hours (H). A deep neural network (DNN) surrogate predicts these indicators from 12 morphological factors, and the optimizer comparison is conducted on the most accurate surrogate identified from a 500/1000/1500/2000-sample scale study. On the selected 2000-sample surrogate, NSGA-II attains stronger objective summaries, the largest Hypervolume, and the lowest IGD, while all three DDPG scenarios remain weaker under post-hoc preference-aligned utility selection. A supplementary checkpoint-sensitivity audit further shows that two surrogates with similar training-fit metrics can still exhibit different extrapolative bound-violation rates, and a same-budget CMA-ES follow-up indicates that surrogate-corner exploitability is broader than DDPG instability alone. A representative small-batch physical probe then adds bounded external grounding: EUIt is the strongest-aligned quantity, the current EG check is weaker but directionally stable, and a revised windowsill-style direct-sun-hours check now brings H into the same broad range as the surrogate values, albeit with weaker stability than EUIt. Overall, the results are best interpreted as evidence of surrogate-conditioned benchmark fragility in urban energy design, not as support for a universally superior policy-learning optimizer.

Keywords: Multi-objective optimization, Urban morphology, Energy performance, Deep reinforcement learning

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Abbreviations and Symbols

Abbreviations	
UM	Urban morphology
EP	Energy performance
UMF	Urban morphological factor
EPI	Energy performance indicator
DRL	Deep reinforcement learning
DNN	Deep neural network
DDPG	Deep Deterministic Policy Gradient
MOO	Multi-objective optimization
HV	Hypervolume
IGD	Inverted Generational Distance

Main symbols	
EUIt	Energy use intensity (kWh/m^2/y)
EG	Energy generation (10^6 kWh/y)
H	Sunlight hours (h)
FAR	Floor area ratio
BD	Building density
OSR	Open space ratio
SVF	Sky view factor
AF	Average floors
AR$_{e-w}$	East-west aspect ratio
AR$_{n-s}$	North-south aspect ratio
OSLI	Open Space Location Index
θ	Block orientation angle (deg)
w_1, w_2, w_3	Reward weights for EUIt, EG, and H
γ, τ	DDPG discount and soft-update factors

1. Introduction

In the context of carbon peaking and carbon neutrality, the decarbonization of urban energy systems has become a central problem in sustainable development [1, 2]. Because cities concentrate both energy demand and renewable-energy opportunities, urban morphology (UM) strongly influences building energy performance (EP), solar access, and environmental quality [3, 4, 6, 5, 22]. In practice, however, urban design and energy planning are still often treated as separate tasks, which weakens block-scale energy performance and limits the effective use of renewable resources [7, 8]. Developing an integrated framework that links morphology generation, performance evaluation, and design optimization is therefore an important research problem at the intersection of architecture, urban planning, and energy systems [10].

This context also highlights the methodological importance of surrogate-based design studies. Large comparative sweeps over urban-energy design strategies are often bottlenecked by expensive simulation workflows, so surrogate-assisted optimization is increasingly used as a screening layer before full physical validation. Under these conditions, optimizer reliability becomes part of the energy-design problem itself: if a learning-based method is unstable, checkpoint-sensitive, or difficult to benchmark fairly, its practical value for urban energy design remains limited.

1.1. Research Motivation

To address this challenge, computational design methods based on parametric modeling and multi-objective optimization (MOO) have become mainstream [11, 12]. In this line of research, evolutionary algorithms such as NSGA-II are widely used to explore trade-offs among conflicting objectives including energy demand [13], photovoltaic potential [14], and daylight performance [15]. These studies have demonstrated the value of optimization-led urban design, but they also reinforce a common workflow in which the main analytical product is a Pareto archive. For design practice, however, a Pareto archive alone does not resolve whether a method can support preference-conditioned navigation, whether its search behaviour remains stable under repeated runs, or whether its benchmark advantage is robust to surrogate selection.

The core motivation of this study is therefore not to assume that policy learning should outperform Pareto search, but to test a more specific question: can a preference-conditioned policy-learning formulation identify useful high-performing regions of the same surrogate landscape while preserving interpretable morphology patterns under a shared optimization budget? Framed in this way, the study addresses both a design question and a methodological question, namely whether policy learning exposes a useful preference-conditioned search bias for urban morphology design and how that bias should be interpreted relative to a fair NSGA-II benchmark.

1.2. Literature Review and Research Gap

Reviews of simulation-based optimization have established the methodological foundations of the field, but they also delimit the present gap. Nguyen et al. [16] and Machairas et al. [17] review building-performance optimization algorithms at a general level, yet they do not address how a policy-learning optimizer should be evaluated against a Pareto-search baseline in an urban morphology setting. More recent reviews by Pan et al. [20] and Westermann and Evins [23] clarify the computational value of simulation acceleration and surrogate modelling, but they stop short of asking how surrogate fidelity affects the credibility of optimizer ranking. Zhao et al. [21] further demonstrate coordinated surrogate-assisted optimization of building and microclimate performance, but their study does not examine whether the optimization method itself remains stable, preference-sensitive, and fairly benchmarked under a shared surrogate.

At the application level, existing urban-form optimization studies have demonstrated important capabilities, but each leaves part of the present problem unresolved. Ascione et al. [13] optimize envelope design, Li et al. [14] target photovoltaic-oriented inverse urban design, and Liu et al. [15] optimize urban block forms for daylight, energy use, and photovoltaic potential. Recent large-scale morphology-energy studies have also shown that GIS-informed deep learning can reveal how urban spatial structure shapes building energy demand across broad urban areas and under extreme climate events [5, 22, 9]. These studies strengthen the empirical case that morphology matters, but they still do not resolve whether optimizer conclusions remain stable, fairly benchmarked, and transferable across surrogate choices. Similarly, recent studies using NSGA-II or related evolutionary schemes in energy-oriented urban design [18, 19, 25, 26, 27, 28] provide stronger objective trade-offs, shading designs, or envelope-shape combinations, yet they do not establish whether designers can obtain a reusable policy for navigating the same design space, nor do they isolate how much of the reported advantage depends on the selected surrogate or benchmark protocol.

The reinforcement-learning literature does not close this gap either. General DRL surveys [29] and continuous-control benchmarks in robotics, games, language, and autonomous driving [30, 31, 32, 33] confirm that policy learning is effective in high-dimensional decision problems, while DQN [34] and DDPG [35] define the algorithmic basis for discrete and continuous control. However, these studies are not concerned with urban morphology design, surrogate-dependent benchmark fairness, or morphology-level interpretability. In the energy domain, DRL has been increasingly adopted for building HVAC control [36, 37] and power-system operation [38, 39], but these applications address operational control rather than static morphology generation. Recent *Applied Energy* studies already benchmark stronger actor-critic baselines such as SAC, TD3, PPO, and TRPO in continuous-control energy settings and report that SAC is often among the strongest or most data-efficient candidates [40, 41, 42]. *Applied Energy* has also begun to report CMA-ES-based optimization in neighboring energy-system applications [43]. Accordingly, the present paper should be interpreted as an intentionally narrow DDPG-versus-NSGA-II study within one surrogate-conditioned urban-design workflow, not as a claim that the compared pair exhausts the relevant optimizer space. In particular, SAC and other entropy-regularized actor-critic variants are discussed here only as relevant external baselines from the literature; they are not executed in the current work and therefore should not be folded into the paper's causal claims about deterministic policy-gradient behavior. Against that backdrop, the literature still lacks a clear answer to a more specific scientific question: under a shared surrogate environment for block-scale urban morphology, how fragile are optimizer rankings to surrogate selection, and what can a bounded DDPG-versus-NSGA-II comparison still reveal about morphology-level search bias?

This unresolved question defines the knowledge gap addressed in this paper. The gap is not the absence of optimization, surrogate modelling, or DRL in isolation; those elements already exist in the literature. Rather, the missing contribution is a benchmarked study that places a continuous-control DRL agent and NSGA-II in the same surrogate environment, evaluates them under matched budgets, and interprets the outcome through optimizer-ranking fragility, morphology-level fingerprints, and the dependence of benchmark conclusions on surrogate selection.

1.3. Study Contributions and Paper Structure

Against this background, this study develops and evaluates a DRL-DNN integrated framework for block-scale urban morphology design. Three specialist agents with distinct reward preferences are trained and compared with an NSGA-II benchmark under the same surrogate environment.

1. This study formulates block-scale urban morphology optimization as a reward-weighted policy-learning task and embeds that task in a shared DNN surrogate environment for EUIt, EG, and H.

2. This study shows that, on the selected highest-accuracy surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in archive-level Pareto metrics and preference-aligned utility, while the DDPG runs retain substantial late-stage instability.
3. This study interprets the resulting comparison as a surrogate-conditioned benchmark-fragility problem rather than as evidence of optimizer superiority, and it uses checkpoint-sensitivity analysis, a CMA-ES follow-up, and independent candidate re-evaluation to bound the credibility of surrogate-space ranking.
4. This study combines objective-space analysis with morphology-level interpretation, nonlinear response diagnostics, and a bounded representative physical probe to identify which observed method differences are descriptive morphology signatures and which claims should remain explicitly conditional on the selected surrogate and validation mode.

The remainder of this paper is organized as follows. Section 2 presents the DRL-DNN integrated framework, including parametric modeling, performance simulation, surrogate modeling, and the DDPG optimization engine. Section 3 reports the experimental results and comparative analysis. Section 4 discusses the implications and limitations of the study. Section 5 concludes the paper. Additional case-setting and surrogate-validation details are provided in Appendix A, and benchmark-robustness diagnostics are reported in Appendix B.

2. Methodology

This study proposes an integrated surrogate-assisted workflow for block-scale urban morphology design. As illustrated in Fig. 1, the framework combines parametric morphology generation, performance evaluation, surrogate modeling, and policy learning. A parametric dataset is first constructed to link urban morphological factors (UMFs) with key energy performance indicators (EPIs). A DNN surrogate is then trained for rapid performance prediction, after which a DDPG agent interacts with the surrogate environment to search for morphology strategies that balance energy demand, energy generation, and sunlight access.

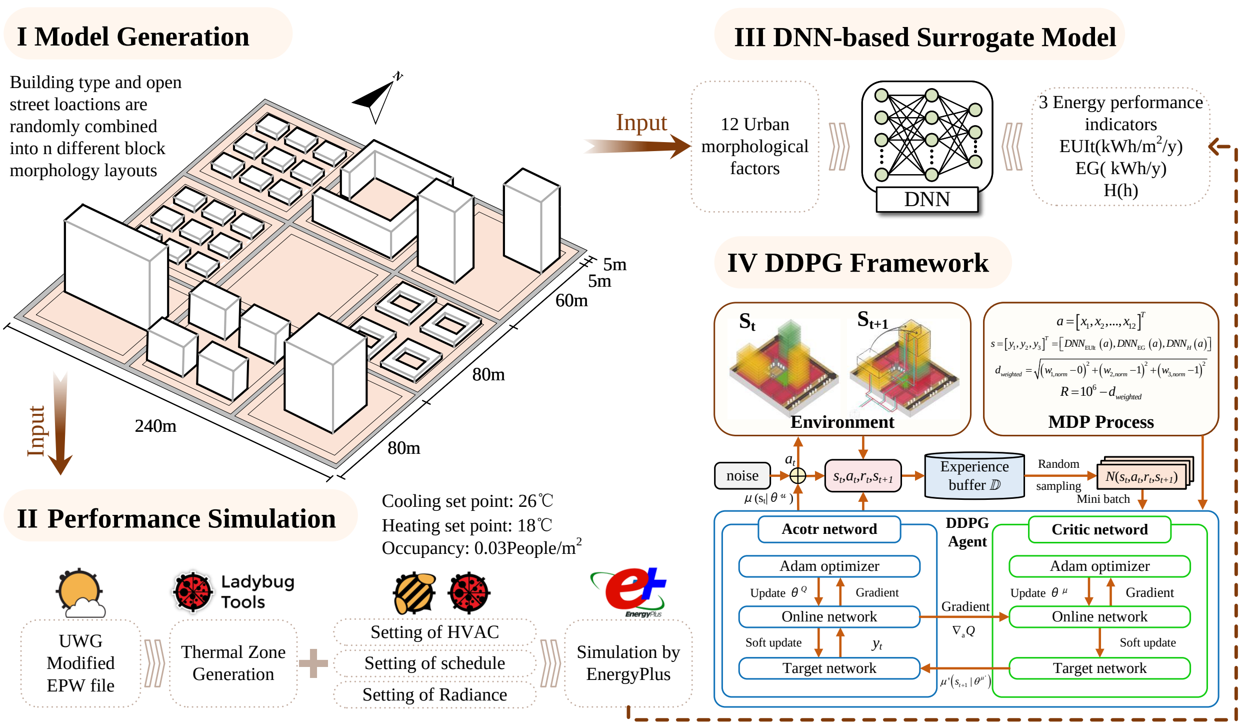

Figure 1: Overview of the proposed method. The left block summarizes parametric morphology generation and performance simulation, while the right block shows the surrogate-assisted optimization workflow used during policy learning.

2.1. Case Study and Parametric Modeling

The selected research area is Jianhu County, located in Jiangsu Province in the eastern coastal region of China. The county is situated between 33°16′ and 33°41′ N latitude and 119°33′ and 120°05′ E longitude, with an average elevation of 2 m. It has an administrative area of approximately 1160 km^2 and a population of around 600,000, resulting in a population density of about 520 persons/km^2. According to the Chinese Code for Thermal Design of Civil Buildings, Jianhu lies within the hot-summer and cold-winter climate zone and is classified as having a humid subtropical climate under the Koppen-Geiger system. The climate necessitates both building heating and cooling, as the average dry-bulb temperature is 27.1°C in the hottest month (July) and 1.6°C in the coldest month (January), with an annual average of 15.0°C. Furthermore, the region presents favorable conditions for solar energy utilization, receiving annual solar radiation of about 1250 kWh/m^2 and approximately 1800 hours of sunshine per year.

To generate a dataset that links urban form to energy performance, a parametric modeling approach was implemented on the Grasshopper platform. The core of this approach is a standard urban block subdivided into a 3×3 grid to create nine land units. Based on an analysis of the existing urban fabric, representative residential building prototypes, including point, slab, and courtyard types, were defined. In the parametric model, one land unit is designated as open space, while the remaining eight units are populated with these building prototypes to generate a wide variety of block morphologies.

This parametric generation process is controlled by twelve selected UMFs. These include Floor Area Ratio (FAR), Building Density (BD), Open Space Ratio (OSR), Average Floors (AF), Setback Diversity (SD), Shape Coefficient (SC), Average Perimeter-to-Area Ratio (PAR), the east-west and north-south aspect ratios (AR_{e-w} and AR_{n-s}), average Sky View Factor (SVF), Open Space Location Index (OSLI), and Block Orientation Angle θ. Once the geometric model of the block is configured on the Grasshopper platform, these UMFs can be calculated using its built-in tools. A dataset-scale study was carried out over 500, 1000, 1500, and 2000 morphologies, and the final optimizer comparison uses the most accurate surrogate identified from that study. The formulas used in this work are summarized in Table 1.

Table 1: Calculation formulas of the morphological factors.

Factor	Symbol	Formula
Floor area ratio	FAR	$\mathrm{FAR} = \dfrac{\sum_{i=1}^{n} S_i}{A_s}$
Setback diversity	SD	$\mathrm{SD} = h_{\max} - h_a$
Average floors	AF	$\mathrm{AF} = \dfrac{\sum_{i=1}^{n} S_i}{\sum_{i=1}^{n} f_i}$
East-west aspect ratio	AR_{e-w}	$\mathrm{AR}_{e-w} = \dfrac{1}{n} \sum_{i=1}^{n} \dfrac{H_i}{W_{i,e-w}}$
North-south aspect ratio	AR_{n-s}	$\mathrm{AR}_{n-s} = \dfrac{1}{n} \sum_{i=1}^{n} \dfrac{H_i}{W_{i,n-s}}$
Sky view factor	SVF	$\mathrm{SVF} = \dfrac{1}{2\pi} \int_0^{2\pi} \cos^2(\beta(R,\theta))\, d\theta$
Building density	BD	$\mathrm{BD} = \dfrac{\sum_{i=1}^{n} f_i}{A_s}$
Open space ratio	OSR	$\mathrm{OSR} = \dfrac{A_s - \sum_{i=1}^{n} f_i}{\sum_{i=1}^{n} S_i}$
Shape coefficient	SC	$\mathrm{SC} = \dfrac{\sum_{i=1}^{n} c_i}{\sum_{i=1}^{n} h_i f_i}$
Perimeter-to-area ratio	PAR	$\mathrm{PAR} = \dfrac{\sum_{i=1}^{n} p_i}{\sum_{i=1}^{n} f_i}$

Note: n is the number of buildings; S_i is the total floor area of building i; A_s is the block area; f_i is the building footprint area of building i; $h_{\max}$ and h_a are the maximum and average building heights of the block; H_i is the building height; $W_{i,e-w}$ and $W_{i,n-s}$ are the east-west and north-south canyon widths associated with building i; c_i is the surface area of building i; p_i is the perimeter of building i; and β is the angle from the center point to the maximum obstacle height at a distance equal to the constant search radius R.

In addition to the factors in Table 1, OSLI is introduced to characterize the topological position of the open space within the 3×3 grid and to account for its influence on local wind, thermal, and daylight conditions [19, 20]. OSLI is an integer variable ranging from 0 to 8. Furthermore, the block orientation angle θ, ranging from $[-45°, 45°]$, is included to regulate solar access. Additional geometric details of the block layout and the representative building prototypes are provided in Appendix A.

2.2. *Performance Simulation and Optimization Objectives*

This study evaluates the energy behavior and solar potential of residential blocks using three key EPIs: total Energy Use Intensity (EUIt), Energy Generation (EG), and Sunlight Hours (H). Together, these indicators represent the demand, supply, and daylight dimensions of block-scale energy performance.

The three targets retain the same physical interpretation as in the original Ladybug/Honeybee/EnergyPlus workflow, but the dataset used in the present study was generated with the documented fallback analytic simulator rather than direct full-stack simulation calls. In the synced dataset metadata, this is recorded as a fallback analytic simulation mode. The evidence in this study should therefore be interpreted as surrogate-assisted optimization on an analytically reconstructed response surface that preserves the intended demand-supply-daylight trade-offs within the sampled design domain.

Under this response definition, H denotes daylight access over the winter design window represented in the original workflow by south-facing ground-floor windowsill test points on a typical coldest day (January 20, 08:00–16:00). In the present dataset, the fallback simulator approximates this daylight response from the morphological factors while preserving the same directional interpretation: higher H indicates better solar access.

EUIt remains the energy-demand target of the framework and corresponds conceptually to annual residential energy use under the envelope, internal-load, and operating assumptions of the original Honeybee/EnergyPlus model. In the present dataset, the fallback simulator approximates this demand response from the same morphology descriptors rather than from direct annual EnergyPlus execution.

EG remains the renewable-generation target and corresponds conceptually to annual rooftop PV potential under the standardized assumptions of the original Ladybug-Radiance workflow. In the present dataset, the fallback simulator approximates this generation response from the morphology descriptors while preserving the same higher-is-better interpretation.

Table 2: Reference assumptions of the original physics-based building-energy workflow. The results reported in this manuscript are generated from the documented fallback analytic simulator rather than direct EnergyPlus runs.

Classification	Parameter	Specification
System settings	Weather data	EPW files modified based on UWG urban microclimate simulations
System settings	Simulation period	Annual simulation (January 1 to December 31)
Envelope thermal performance	Wall U-value	0.8 W/(m^2·K)
Envelope thermal performance	Roof U-value	0.5 W/(m^2·K)
Envelope thermal performance	Floor U-value	1.5 W/(m^2·K)
Envelope thermal performance	Window properties	U-value = 2.70 W/(m^2·K); Solar Heat Gain Coefficient (SHGC) = 0.78
Window-to-wall ratio	North & South facades	0.4
Window-to-wall ratio	East & West facades	0.1
Internal loads	Occupant density	0.03 persons/m^2; metabolic rate = 100 W/person
Internal loads	Lighting power density	5 W/m^2
Internal loads	Equipment power density	1.9 W/m^2
Internal loads	Ventilation rate	30 m^3/(h·person)
HVAC system	Thermostat setpoints	Heating: Dec 1–Feb 28, 18°C; Cooling: Jun 15–Aug 31, 26°C
HVAC system	Infiltration rate	1 ACH (natural air leakage)

2.3. *DNN-based Surrogate Model*

In this study, a DNN is constructed as a predictive surrogate model to estimate the three target performance indicators from the simulated dataset. The model adopts a multi-layer feedforward architecture in which information flows from the input layer through the hidden layers to the output layer. Through stacked fully connected transformations, the network captures the nonlinear relationships between urban morphological factors and block-scale performance outcomes.

The DNN training process consists of two key stages: forward propagation and backpropagation. In the forward stage, the network processes input data to generate a prediction. During backpropagation, the prediction error is propagated backward from the output layer to the input layer. This process involves calculating the gradient of the loss function with respect to the network weights and iteratively updating the model parameters to reduce prediction error:

$$\mathbf{w} = \mathbf{w} + \eta\nabla_{\mathbf{w}} = \mathbf{w} + \eta\,\text{backpropagation}(f(\mathbf{x}))\,, \tag{1}$$

where η is the learning rate and $f(\mathbf{x})$ is the objective function to minimize.

To determine the DNN architecture and training settings, Bayesian Optimization (BO) was used for automated hyperparameter tuning. Compared with grid search, BO guides the search process by constructing a probabilistic model of the objective function and repeatedly selecting promising hyperparameter combinations. After the optimal settings were determined, model performance was evaluated using five-fold cross-validation. Because the surrogate is subsequently used for optimization, it is treated here as an optimization-enabling approximation rather than as a direct replacement for the original simulation workflow.

Table 3: Optimal hyperparameters of the DNN determined by Bayesian Optimization.

Hyperparameter	Value
Number of dense layers	1
Units of 1st layer	128
Units of 2nd layer	–
Dropout rate	0.143
Learning rate	0.00134
Batch size	32
Optimizer	Adam
Activation	ReLU

To ensure a rigorous evaluation of generalization capability and avoid bias from a single train-test split, the final highest-accuracy surrogate was evaluated by five-fold cross-validation over the 2000-sample regime. The coefficient of determination (R^2) and Mean Absolute Error (MAE) were used as key evaluation metrics.

Table 4: Summary of DNN prediction performance using five-fold cross-validation.

Objective	Mean R^2	Mean MAE	Std. R^2	Std. MAE
EUIt	0.983	0.209	±0.001	±0.005
EG	0.972	0.026	±0.002	±0.001
H	0.992	0.025	±0.001	±0.001

The evaluation results show that the selected 2000-sample surrogate maintains R^2 values close to 1 for all three objectives while keeping MAE low relative to the observed target ranges. In the scale study, this model achieves a mean target nMAE of 0.0182 and a mean tail nMAE of 0.0203, which together indicate the strongest predictive accuracy among the tested candidates. Taken together, these results justify using the selected 2000-sample surrogate as the basis for the final optimizer comparison.

2.4. DDPG-based Optimization Framework

The parametric modeling process and the DNN surrogate together provide the basis for the integrated optimization framework linking urban morphology and energy performance. To optimize these indicators jointly, a DRL framework was developed, as shown in Fig. 2. The framework has two components. The first defines the optimization problem through the environment, action space, state representation, and reward. The second implements the DDPG agent, which interacts with the environment through an experience replay buffer and updates its policy iteratively.

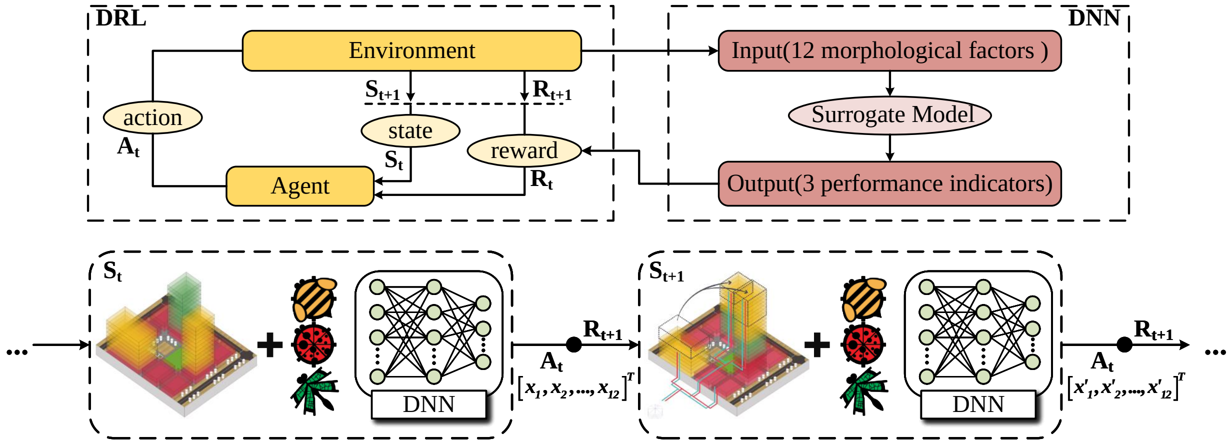

Figure 2: Conceptual framework of the DRL-based optimization. The upper part summarizes the relationship between DRL, the environment, and the DNN surrogate; the lower part details the stepwise Markov Decision Process used by the actor to query the surrogate and receive updated state and reward signals.

2.4.1. Problem Formulation

To address the MOO problem in urban energy performance, this study formulates the optimization process as a Markov Decision Process (MDP), typically defined by the four-tuple (S, A, P, R). The DRL agent continuously receives state and reward signals from the environment and updates its policy iteratively to improve the morphology solution.

Action space. In the DRL framework, an action is the fundamental unit of interaction between the agent and the environment. In this study, an action is defined as an adjustment to a set of preset building design parameters. Specifically, the action vector is formed by twelve key design parameters:

$$\mathbf{a} = [x_1, x_2, \ldots, x_{12}]^{\mathrm{T}}, \tag{2}$$

where $\mathbf{a}$ represents a specific action selected by the agent and x_i is the selected value for the ith design parameter.

State space. The state is defined as the vector of the key EPIs predicted by the DNN-based surrogate model:

$$\mathbf{s} = [y_1, y_2, y_3]^{\mathrm{T}} = [\mathrm{DNN}_{\mathrm{EUIt}}(\mathbf{a}), \mathrm{DNN}_{\mathrm{EG}}(\mathbf{a}), \mathrm{DNN}_{\mathrm{H}}(\mathbf{a})]^{\mathrm{T}}, \tag{3}$$

where y_1, y_2, and y_3 are the predicted values of EUIt, EG, and H, respectively.

State transition probability. The state-transition probability $P(s' \mid s, a)$ defines the probability distribution of the environment transitioning to the next state given the current state and action. In the model-free DRL method employed here, P is not modeled explicitly; instead, the agent learns directly from samples obtained through interaction with the environment. In our framework, for a given action, the DNN surrogate deterministically or near-deterministically outputs the corresponding performance indicators, thereby defining the next state.

Reward function. The reward function provides the scalar feedback signal that guides learning. Because this study simultaneously minimizes EUIt while maximizing EG and H, a normalized weighted-distance formulation is adopted. First, the three objectives are mapped to a common [0, 1] interval:

$$\begin{aligned} y_{1,\mathrm{norm}} &= \frac{y_1 - y_{1,\min}}{y_{1,\max} - y_{1,\min}}, \\ y_{2,\mathrm{norm}} &= \frac{y_2 - y_{2,\min}}{y_{2,\max} - y_{2,\min}}, \\ y_{3,\mathrm{norm}} &= \frac{y_3 - y_{3,\min}}{y_{3,\max} - y_{3,\min}}. \end{aligned} \tag{4}$$

In the normalized space, the utopian point is defined as [0, 1, 1], representing the ideal state in which all objectives simultaneously reach their best values. The weighted Euclidean distance between the current state and the utopian point is then calculated as

$$d_{\text{weighted}} = \sqrt{(w_1 y_{1,\text{norm}} - 0)^2 + (w_2 y_{2,\text{norm}} - 1)^2 + (w_3 y_{3,\text{norm}} - 1)^2}, \tag{5}$$

where w_1, w_2, and w_3 are preset non-negative weight coefficients whose sum equals 1. In this study, the normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. This choice makes the reward scale dependent on dataset coverage, and therefore the bound selection must be interpreted as a practical normalization strategy rather than a theoretically unique definition. Finally, the reward function is defined as

$$R = 10^6 - d_{\text{weighted}}. \tag{6}$$

The reward function therefore transforms the multi-objective design problem into a scalar optimization target that still preserves design preferences through the weight coefficients. The factor 10^6 is used only as a numerical offset to keep rewards positive and to stabilize optimization logs; it does not alter the ranking induced by the weighted distance.

2.4.2. DDPG Agent Implementation

To solve the formulated MDP, this study adopts the DDPG algorithm. DDPG is a model-free actor-critic method well suited to high-dimensional continuous action spaces. It concurrently trains an actor network, which selects actions, and a critic network, which evaluates the value of those actions under the reward function. Here, the focus is on the implementation used for urban morphology optimization, while additional convergence and benchmark details are reported in Appendix B.

The implementation is structured around a multi-scenario analysis designed to investigate the agent's ability to generate targeted design solutions based on varying decision-maker preferences. Multiple independent DDPG agents are trained, each guided by a unique reward function reflecting a specific design priority. The common architectural hyperparameters are summarized in Table 5.

Table 5: Hyperparameter settings for the DDPG agent.

Parameter	Value
Reward discount factor (γ)	0.999
Learning rate (actor)	0.001
Learning rate (critic)	0.002
Layers (actor network)	2
Units per layer (actor)	64, 32
Layers (critic network)	2
Units per layer (critic)	64, 32
Soft update factor (τ)	0.001
Batch size	128
Replay buffer size	1×10^6
Maximum episodes	600
Max steps per episode	40
Noise decay factor	0.9998

The actor and critic networks both use two hidden layers with 64 and 32 neurons. This relatively simple architecture was found to be effective for the problem complexity while avoiding the instability often associated with deeper networks. The critic learning rate is slightly larger than the actor learning rate, allowing the value function to stabilize faster and provide a reliable training signal to the policy network. The discount factor is set to 0.999, encouraging farsighted strategies focused on long-term cumulative rewards.

To ensure adequate exploration, Gaussian noise with an initial variance of 1.0 is added to the actor outputs and gradually reduced using a decay factor of 0.9998. A replay buffer with a capacity of one million transitions is used to decorrelate samples and stabilize learning; at each training step, a mini-batch of 128 transitions is sampled to update the networks.

The central pillar of the implementation is the configuration of three distinct optimization scenarios. These scenarios are realized by adjusting the weight coefficients (w_1, w_2, w_3) in the reward function:

- **Scenario 1: Balanced performance.** Equal weights are assigned, $w_1 = w_2 = w_3 = \frac{1}{3}$, producing an unbiased preference among minimizing EUIt, maximizing EG, and maximizing H.
- **Scenario 2: Energy saving focus.** The weights are set to $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$, prioritizing reduced energy consumption.
- **Scenario 3: Energy generation focus.** The weights are set to $(w_1, w_2, w_3) = (0.2, 0.6, 0.2)$, prioritizing renewable energy production.

By training a separate agent for each of these scenarios, the DRL framework exposes how different reward scalarizations bias search within the same surrogate environment. The overall prediction distributions remain tightly concentrated around the one-to-one line for all three objectives, while the residual distributions in Fig. 5 indicate that most prediction errors remain within the $\pm 2\sigma$ band. Even so, these error statistics should be interpreted together with the extreme-region diagnostics and candidate re-evaluation results, because optimization concentrates attention on the parts of the response surface where surrogate misspecification is most consequential.

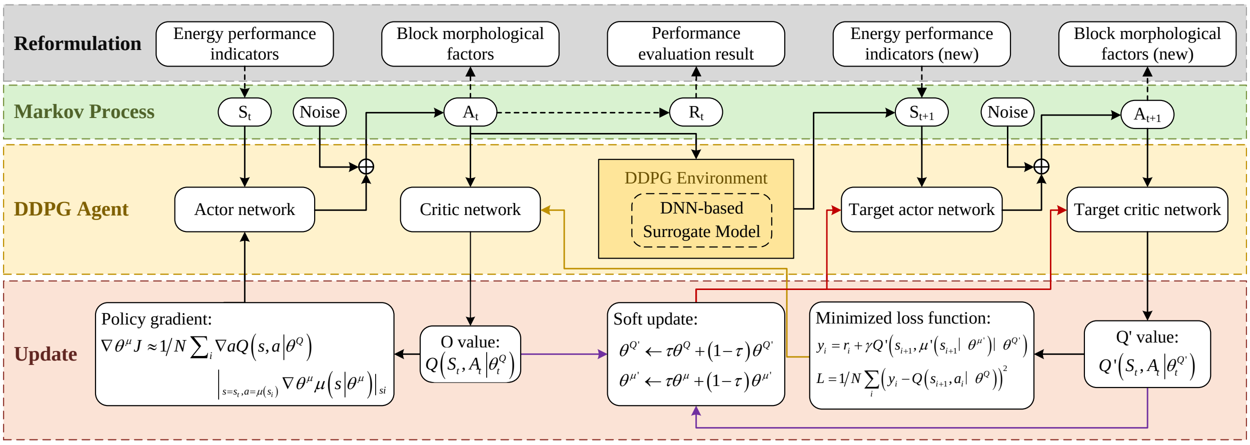

Figure 3: Architecture of the DDPG agent, showing the actor-critic structure and learning process.

3. Results

3.1. Validation Boundary

Before interpreting optimizer differences, it is necessary to state the current validation boundary explicitly. The benchmark comparison itself is still conducted on the selected 2000-sample surrogate, and the imported publication artifact remains tagged as `fallback_analytic`. However, the present revision now includes a small-batch physical-stack probe on two representative candidates, which provides a more informative hierarchy of evidence than was available in earlier rounds.

The resulting hierarchy is asymmetric across the three objectives. EUIt is currently the strongest-aligned quantity: the representative DDPG and NSGA-II candidates yield physical EUIt values of 74.379 and 74.057 kWh/m^2/y, compared with surrogate values of 66.000 and 69.443 kWh/m^2/y. The EG check is weaker but still directionally stable under the current rooftop-PV proxy. The revised H check, based on a windowsill-style direct-sun-hours calculation over the January 20 design window, now yields physical values of 7.525 and 6.075 against surrogate values of 7.850 and 7.569. This is materially stronger than the earlier point-illuminance proxy, but it still remains a small-batch physical check rather than a publication-grade full-stack rerun. Accordingly, the benchmark comparison below should be read as strongest for surrogate-space ranking and EUIt-oriented interpretation, weaker for EG, and still more limited on the H side than on EUIt.

3.2. Surrogate Model Performance

Based on the simulations generated through the Grasshopper process, a surrogate scale study was carried out over 500, 1000, 1500, and 2000 simulated morphologies to identify the most accurate model available for this study. The final optimizer comparison uses the 2000-sample DNN with standard scaling on both inputs and targets and a single hidden layer with 128 units. To reduce overfitting and training instability during candidate selection, early stopping and dropout were employed throughout the search.

As summarized in Tables 3 and 4, the final DNN achieved high predictive accuracy and strong generalization on the expanded 2000-sample regime. Appendix A further reports RMSE values of 0.279 for EUIt, 0.033 for EG, and 0.032 for H, with normalized RMSE values below 3.1% for all three targets. The parity plots in Fig. 4 show that the predictions remain tightly concentrated around the one-to-one line, while the residual distributions in Fig. 5 indicate that most prediction errors remain within the $\pm 2\sigma$ band. These statistics should nevertheless be interpreted together with the extreme-region diagnostics in Appendix A and the candidate re-evaluation results in Appendix B, because optimization is most sensitive to misspecification in the regions of the response surface that repeatedly attract the search process.

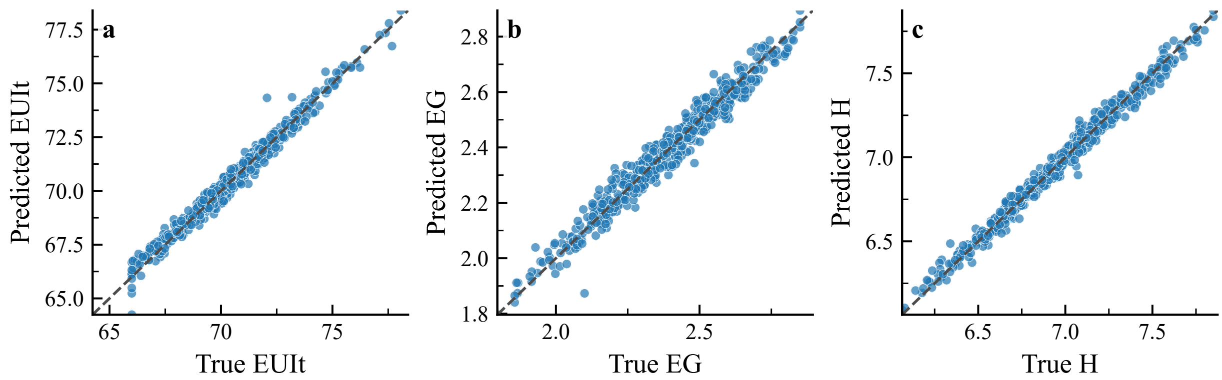

Figure 4: Parity plots comparing surrogate predictions against simulated targets for the three objectives: (a) EUIt, (b) EG, and (c) H. The dashed line denotes perfect agreement.

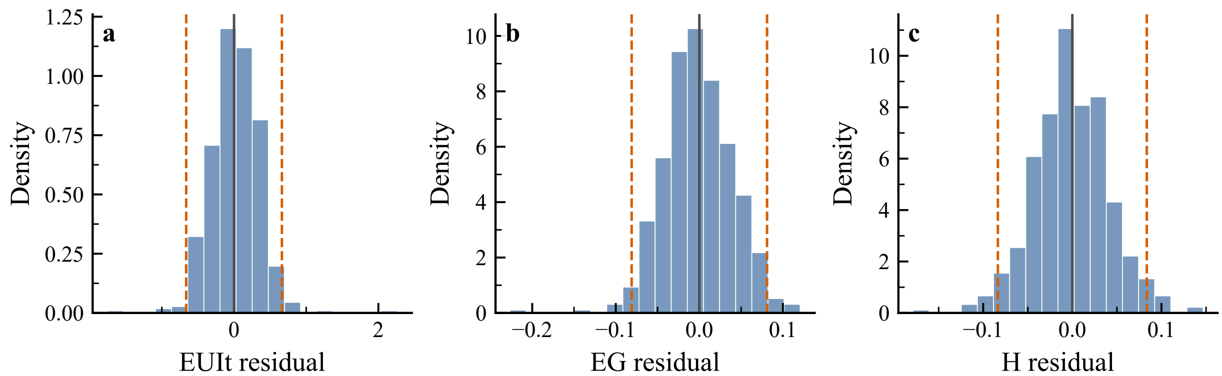

Figure 5: Residual distributions of the surrogate predictions for the three objectives: (a) EUIt, (b) EG, and (c) H. Dashed lines indicate the $\pm 2\sigma$ range around zero residual.

3.3. DDPG Training Dynamics

The learning dynamics of the agents were monitored to assess whether the policy updates approached a recurrent high-reward regime. Figure 6 summarizes the mean learning curves and one-standard-deviation envelopes across the available seeds for the three DDPG scenarios. Each scenario was trained with 20 seeds, and each seed was run for 600 episodes under the guarded surrogate evaluator, which provides a more informative view of training behavior than a single-curve presentation.

The multi-seed curves show that cumulative reward increases sharply during training and then transitions into a narrower high-reward regime, indicating a shift from broad exploration to more concentrated surrogate-region search. On the selected 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401, 320, and 398 for the balanced, saving-focused, and generation-focused scenarios, respectively. The balanced scenario is visibly smoother and exhibits lower cross-seed fluctuation in reward and output trajectories than the other two scenarios, whereas the saving-focused and generation-focused scenarios do not become uniformly less variable across all metrics. At the same time, late-stage regression remains common, with mean best-to-final reward gap ratios around 0.62, 0.71, and 0.71 and substantial-regression fractions of 95%, 85%, and 100% across the three scenarios. These curves should therefore be interpreted as evidence that policy learning can reach high-performing surrogate regions rather than as proof of stable asymptotic convergence or global optimality. The learning-curve evidence is therefore complemented by benchmark comparisons and independent candidate re-evaluation.

The three scenarios remain distinguishable but still overlap in aggregate objective space. In the 20-seed experiments on the selected surrogate, the energy-saving-focused scenario yields the lowest mean EUIt (67.10 kWh/m^2/y), the generation-focused scenario yields the highest mean EG (2.763$\times10^6$ kWh/y), and the balanced scenario retains the highest mean H (7.687 h). These differences should be interpreted as preference-conditioned shifts within one surrogate landscape rather than as proof of sharply separated optima.

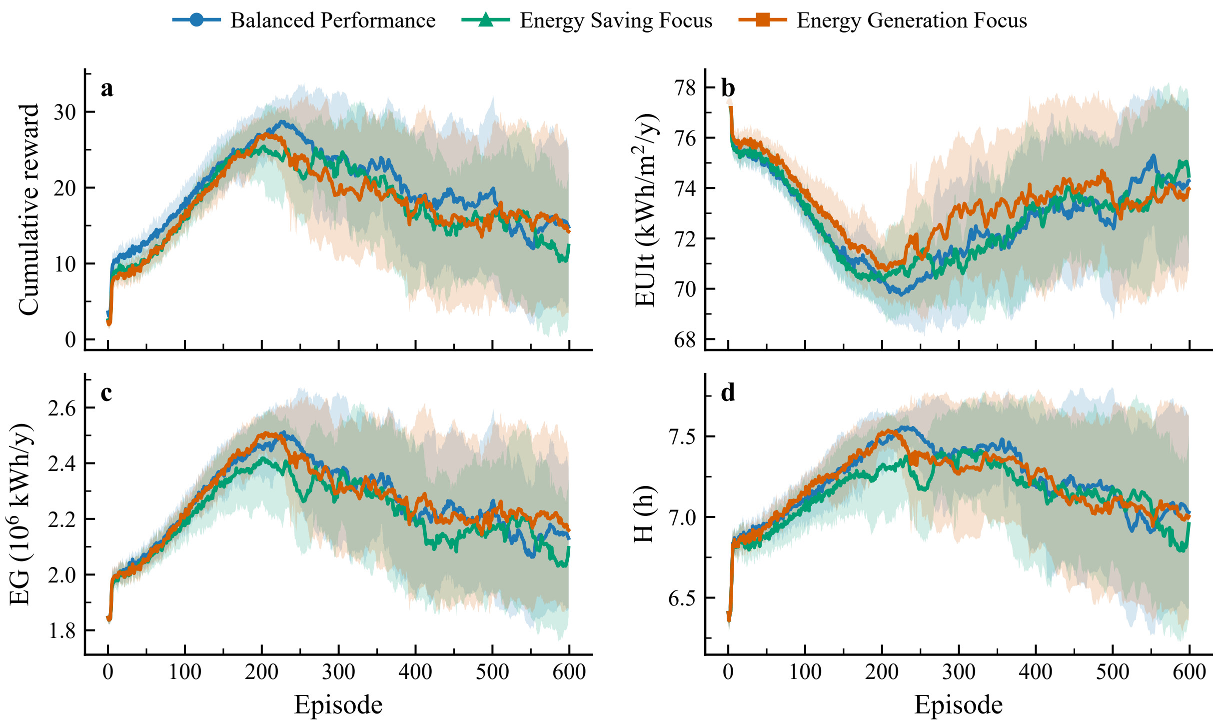

Figure 6: DDPG learning curves during guarded multi-seed training. Solid lines denote mean trajectories across 20 seeds per scenario and shaded regions indicate one standard deviation.

3.4. Multi-objective Optimization Results

Before presenting the comparative results, it is necessary to clarify how the DRL solution clusters were generated. To ensure statistical robustness, each of the three DRL scenarios was trained independently across 20 random seeds.

This produces a distribution of high-performance solutions for each agent, reflecting a recurrent high-performing region visited by its learned policy rather than a single stable convergence point. The analysis below first explores the learned tendencies of the three DRL scenarios and then compares them against the NSGA-II benchmark.

3.4.1. Multi-scenario DRL Behavior

The multi-scenario DRL setup is retained here as a diagnostic rather than as proof of a robust preference-navigation advantage. By specifying a design preference a priori through reward-function weights, the framework trains three reward-weighted agents under the same surrogate environment. Figure 7 illustrates the differences among the scenarios. On the selected surrogate, the three DDPG scenarios still occupy overlapping regions, but the scenario-level objective differences remain readable through their morphology summaries.

The most important observation is that the reward-conditioned DDPG agents produce overlapping but still distinguishable surrogate-space clusters, yet they do not surpass the fair-budget NSGA-II archive on the selected surrogate. As shown in Fig. 7, the three DDPG scenarios form overlapping clusters, whereas NSGA-II reaches better average objective values and stronger Pareto indicators on the same 2000-sample surrogate. In the present paper, these morphology-level differences are therefore treated as descriptive evidence of how reward scalarization biases search on one surrogate checkpoint, not as proof of a robust preference-articulation advantage.

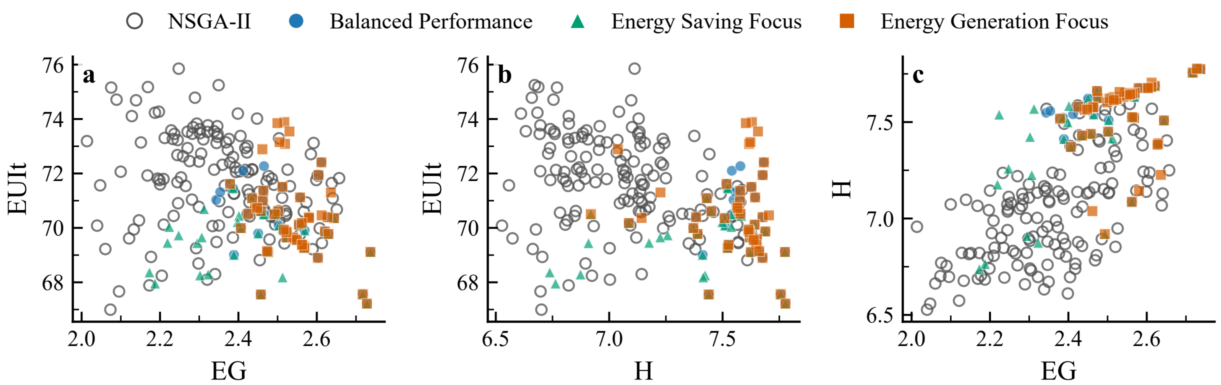

Figure 7: Pairwise objective-space comparison on the selected 2000-sample surrogate. Each point is one candidate solution. The three DDPG scenarios still form overlapping clusters, while the fair-budget NSGA-II archive reaches stronger objective values on the same surrogate.

These objective-space differences become more interpretable in the morphology signatures shown in Fig. 8. The heatmap summarizes relative median UMF levels for each method. On the selected surrogate, the balanced scenario tends toward relatively higher OSR and SVF, the generation-focused scenario shifts toward higher SD and OSLI while retaining high SVF, and the saving-focused scenario combines lower BD with comparatively higher FAR and AF. The contrasts remain moderate and overlapping rather than sharply separated, so they should be read as surrogate-conditional fingerprints rather than as stable design rules.

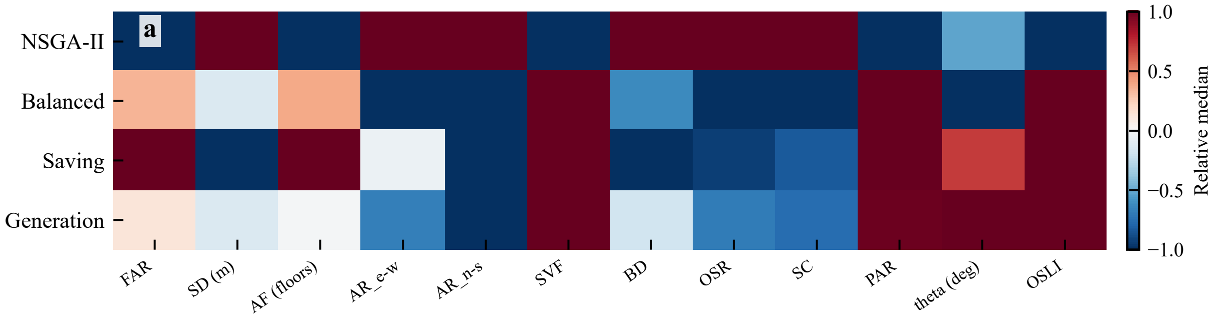

Figure 8: Relative median morphology signatures across methods. Colors encode within-factor contrast across the four method-level median profiles, highlighting which UMFs shift most under the three DDPG reward settings relative to NSGA-II.

3.4.2. Comparison with NSGA-II

After establishing the internal behavior of the DRL agents, we compare them to the NSGA-II benchmark on the selected surrogate from the scale study. Table 7 summarizes this apples-to-apples comparison. On this result set, NSGA-II again attains the strongest Pareto-quality metrics (HV = 1.331, IGD = 0.107), while the three DDPG scenarios remain weaker. The generation-focused DDPG run is the strongest of the three DDPG variants in Pareto-quality terms (HV = 1.011, IGD = 0.220), but it still trails NSGA-II. Figure 9 visualizes the same ordering from two complementary angles: scenario-wise objective summaries and preference-aligned utility.

To make the comparison protocol explicit in the main text, Table 6 reports the directly reproducible budget accounting from the synced configuration and rerun metadata. The published comparison matches the per-seed interaction budget and the number of seeds between DDPG and NSGA-II, and it also makes clear that DDPG depends on the pre-trained surrogate whereas NSGA-II does not require policy training. In addition, a representative local CPU-only audit was carried out on one balanced-scenario seed for DDPG, one fair-budget NSGA-II run, one CMA-ES run, and one random-search run on the same synced surrogate. The resulting wall-clock values should be interpreted as a minimum reproducible reference rather than as a cross-hardware production benchmark.

This result leads to a narrower interpretation of the benchmark comparison. Rather than claiming superiority of DRL over NSGA-II, the current evidence supports the conclusion that DDPG does not match fair-budget NSGA-II on the selected surrogate. Under post-hoc preference-aligned utility selection, the balanced, saving-focused, and generation-focused DDPG candidates all remain below the corresponding best NSGA-II candidate, with utility deltas of approximately −0.267, −0.140, and −0.046, respectively. This conclusion should be interpreted together with four additional caveats. First, the independent re-evaluation reported in Appendix B was also carried out for the 2000-sample comparison, but it still relies on the fallback analytic simulator rather than a full physical stack. Second, the DDPG runs continue to exhibit frequent late-stage reward regression despite reaching high-reward regions mid-training. Third, the optimizer ranking remains checkpoint-sensitive, which means benchmark conclusions should be tied explicitly to the selected surrogate artifact rather than generalized across checkpoints. Appendix B now makes this point more concrete: the imported publication artifact favors DDPG in Hypervolume, whereas the later strict-highest-accuracy and surrogate-rebenchmark checkpoints both favor NSGA-II in both Hypervolume and IGD. Fourth, a same-budget CMA-ES follow-up on the three locally accessible checkpoint contexts also collapses to the same bounded corner solution, which indicates that surrogate exploitability is not unique to DDPG.

Detailed linear correlation maps, joint distributions, and nonlinear response diagnostics are reported in Appendix B rather than in the main text. In the present study, these diagnostics are used only as interpretive support for the morphology-level shifts in Fig. 8; they are not treated as primary evidence for the benchmark comparison.

Table 6: Main-text budget accounting for the selected 2000-sample benchmark comparison.

Item	DDPG	NSGA-II	Interpretation
Per-seed budget	600 episodes × 40 steps	24,000 evaluations	matched surrogate-query budget
Seed count in published comparison	20 per scenario	20	matched seed count in the selected rerun
Scenario structure	3 reward settings	1 archive search	DDPG results are reported per reward-conditioned scenario
Needs pre-trained surrogate at optimization time	yes	yes	both methods are benchmarked on the same selected surrogate artifact
Additional policy-training overhead	yes	no	a methodological asymmetry that should be kept in mind when interpreting efficiency claims
Representative local CPU-only audit	DDPG 99.4 s / 436.6 MB; CMA-ES 0.40 s / 437.9 MB; random 0.34 s / 543.6 MB	NSGA-II 12.5 s / 437.6 MB	local single-run reference on the synced surrogate; not a full cross-hardware production audit

Note: This table reports the directly reproducible accounting available from the current synced configuration and imported benchmark metadata. It makes the query-count symmetry explicit, but it does not replace a hardware-normalized compute audit.

Table 7: Comparison across fair-budget NSGA-II and the three DDPG scenarios on the selected 2000-sample surrogate.

Method	HV	IGD	EUIt	EG (10^6 kWh/y)	H (h)
NSGA-II	1.331	0.107	66.15	2.844	7.836
Balanced DDPG	0.347	0.720	67.90	2.700	7.687
Saving DDPG	0.625	0.549	67.10	2.716	7.487
Generation DDPG	1.011	0.220	68.46	2.763	7.612

Note: The values in this table are computed on the selected 2000-sample surrogate from the scale study. On this surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in both archive-level Pareto metrics and preference-aligned utility.

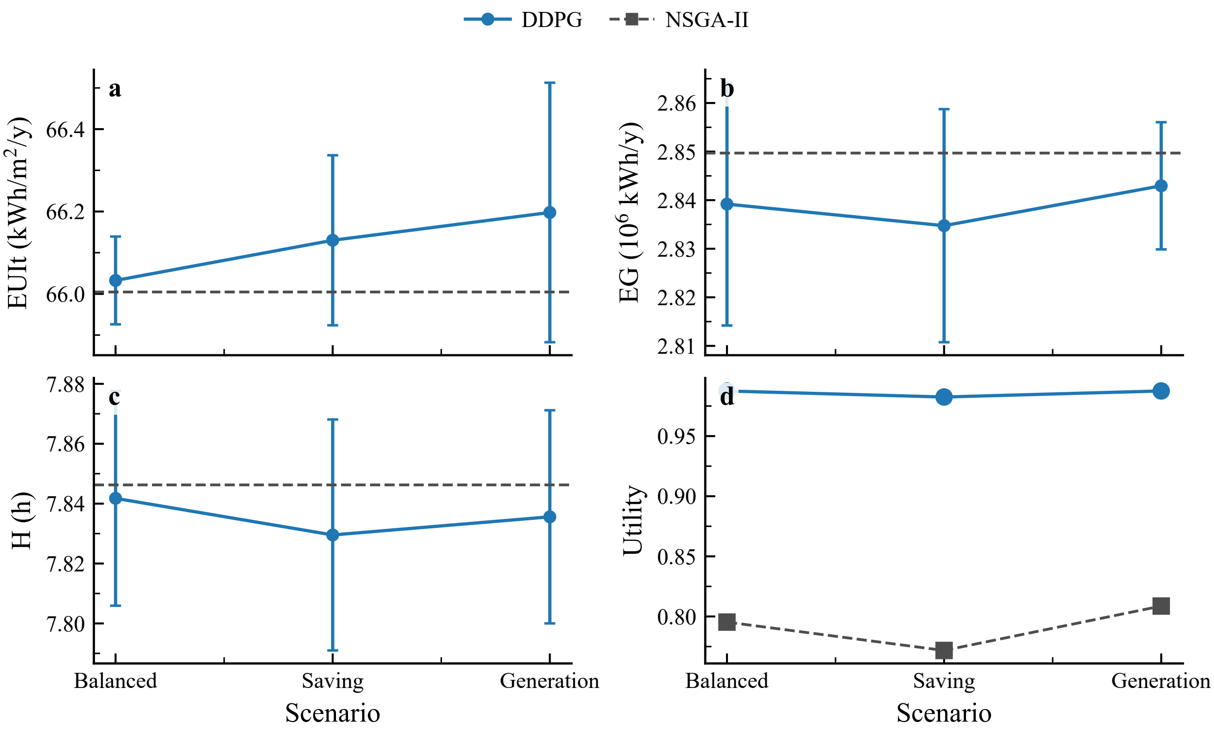

Figure 9: Benchmark summary on the selected 2000-sample surrogate. Panels (a)–(c) compare scenario-wise objective summaries for DDPG, with the NSGA-II mean shown as a dashed reference line. Panel (d) compares preference-aligned utility for DDPG and NSGA-II across the three scenarios.

Figure 10 summarizes the seed-level convergence diagnostics on the selected 2000-sample surrogate. The generation-focused scenario reaches the highest mean best reward, whereas the saving-focused scenario reaches its plateau earliest. However, all three scenarios still show large best-to-final reward gaps and high late-regression fractions, which confirms that the main role of DDPG in the present workflow is to expose checkpoint-specific search bias rather than stable monotonic improvement to a final fixed point.

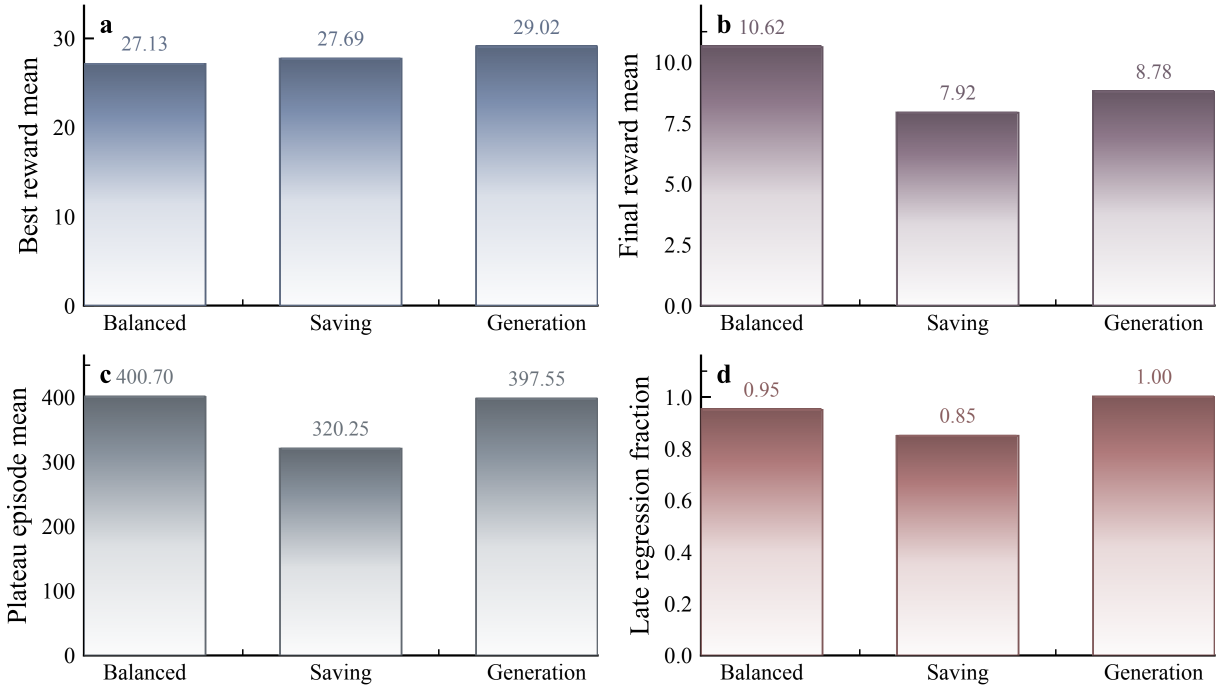

Figure 10: Seed-level convergence diagnostics on the selected 2000-sample surrogate. Panels report the mean best reward, mean final reward, mean plateau episode, and late-regression fraction across 20 seeds per DDPG scenario.

Figure 11 links the optimizer comparison back to the surrogate-selection procedure itself. The regime-winner diagnostics show that surrogate fidelity improves steadily as the dataset grows from 500 to 2000 samples, and that the 2000-sample DNN with standard scaling provides the strongest predictive accuracy among the tested candidates. This figure is included to make the paper's final optimizer benchmark traceable to an explicit surrogate-selection decision.

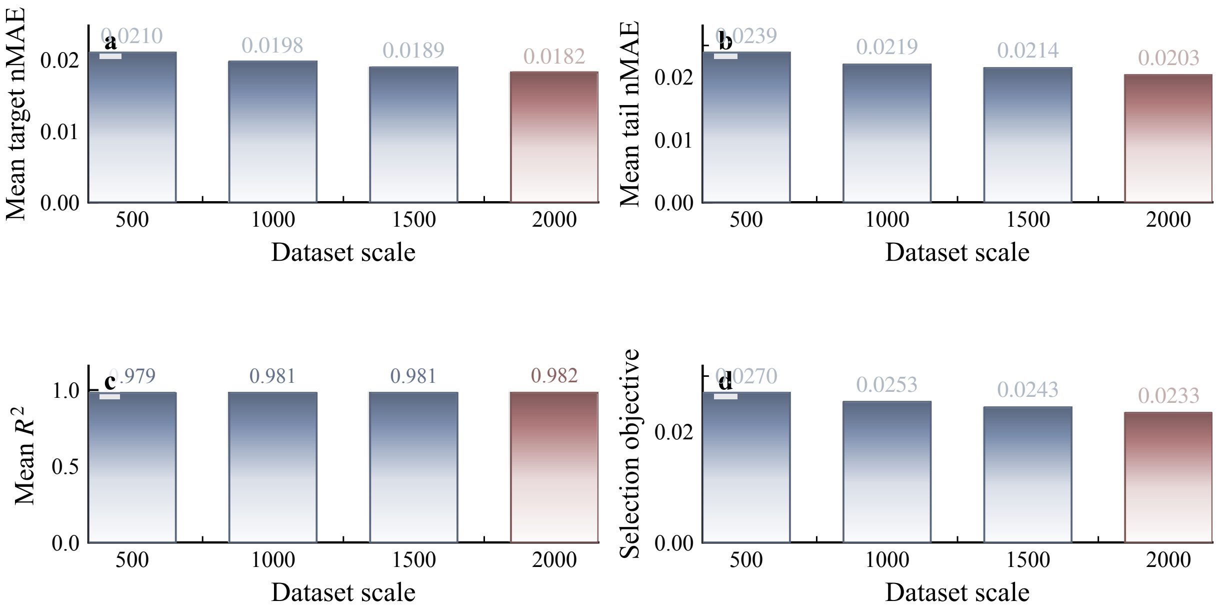

Figure 11: Surrogate scale-study diagnostics across the best regime winner at each dataset scale. Panels report mean target nMAE, mean tail nMAE, mean R^2, and the selection objective from 500 to 2000 samples, with the 2000-sample winner highlighted as the selected model for optimizer comparison.

4. Discussion

This study develops a DRL-DNN integrated framework for block-scale urban morphology design and examines its comparative performance under a surrogate-selection protocol based on predictive accuracy. The analysis indicates that the framework is most credible as a methodological case study of surrogate-conditioned benchmark fragility, rather than as a superior substitute for Pareto-based evolutionary optimization.

The main methodological finding is that reward-conditioned policy learning can produce distinct but overlapping search biases on a given surrogate checkpoint, yet those biases remain less stable and less competitive than the NSGA-II baseline. On the selected 2000-sample surrogate, fair-budget NSGA-II is clearly stronger than all three DDPG scenarios in both objective summaries and Pareto-quality metrics, and it also dominates the post-hoc preference-aligned utility comparison. The three DDPG scenarios remain distinguishable but overlapping in objective space, which means the main contribution of the present comparison is diagnostic: optimizer conclusions remain conditional on surrogate choice, stability, and post-hoc re-evaluation rather than on nominal surrogate-space optima alone.

Even so, the multi-scenario design still reveals how reward scalarization biases the search toward different surrogate-space regions. On the selected surrogate, the resulting differences are clearer in morphology descriptors than in top-line objective means, but they should be interpreted as checkpoint-specific diagnostics rather than as evidence that the policy-learning setup reliably supports design preference articulation across surrogates.

The framework still provides interpretable diagnostic signals. The correlation analysis and the additional nonlinear diagnostics indicate that openness-related variables such as OSR and SVF, together with density-related variables such as FAR and BD, play a dominant role in shaping the objective responses. At the same time, the newly added CMA-ES follow-up in Appendix B shows that a strong continuous optimizer can also collapse to the same surrogate corner solution across all three scalarization settings when the surrogate admits an exploitable plateau. Thus, even though DDPG does not outperform NSGA-II on the selected surrogate, the combined optimizer comparison and morphology analysis remain useful for distinguishing optimizer-specific behavior from broader surrogate-level fragility.

Independent candidate re-evaluation further sharpens this interpretation. On the 2000-sample surrogate, the three DDPG winners do not all drift in the same direction: the balanced candidate improves in EUIt but loses EG and H, whereas the saving-focused and generation-focused candidates drift upward in EUIt and downward in EG after reevaluation through the independent fallback simulation path. The selected NSGA-II candidate also drifts away from its surrogate optimum, but it remains competitive in aggregate objective value. These results reinforce the need to keep optimizer claims bounded by surrogate reliability rather than by surrogate-space optimum values alone.

A limited physical-stack probe now adds a second layer to that conclusion. Under the current best-aligned project-level physical-stack settings, two representative candidates from the balanced DDPG and NSGA-II result sets both return physical EUIt values near 74 kWh/m^2/y, which is above but still close to their surrogate values of 66–69 kWh/m^2/y. The corresponding EG proxy remains directionally similar, and the revised windowsill-style H check now also returns values in the same broad range as the surrogate sunlight-hours values, although with weaker agreement than EUIt. A broader four-method small-batch check further shows that, once the same physical probe is applied to representative DDPG, NSGA-II, CMA-ES, and RandomSearch candidates, the physical EUIt values collapse into a much narrower band than their surrogate-space spread. This shrinkage should not be overinterpreted as proof that the physical design space is optimizer-agnostic. The current data are also consistent with a more limited explanation, namely that the surrogate differentiates methods more strongly than the present small-batch physical projection does. Thus, the present physical evidence is already informative for EUIt, weaker but still structurally useful for EG, and meaningfully improved on the H side even though it still falls short of a full publication-pipeline reconstruction.

From a deployment perspective, the present evidence also supports a simple surrogate self-assessment framework before comparing optimizers on a new urban-energy surrogate. First, check whether nominally similar checkpoints exhibit materially different out-of-bounds rates on random samples from the observed feature ranges. Second, compare optimizer ordering across more than one checkpoint whenever possible, rather than trusting a single selected surrogate. Third, reevaluate any apparent corner solution through an independent deterministic call before treating it as a design recommendation. Fourth, if one optimizer repeatedly converges to a clipped corner while random search does not, treat that region as a likely surrogate-structure warning sign rather than as immediate evidence of design superiority. These checks do not replace physical validation, but they provide a practical screening layer for identifying "optimizer-risky" surrogates in the absence of a full physical stack.

Several limitations should be acknowledged. First, the findings are derived from a single case study in one climate zone. Second, even with 2000 simulated cases, the surrogate still approximates a 12-dimensional continuous design space and therefore remains vulnerable in sparse or extreme regions; accordingly, the current Pareto claims should be interpreted as applying to the interpolated surrogate region rather than to unconstrained extrapolation beyond the sampled design domain. Third, the current comparison remains surrogate-based during optimization, and the independent re-evaluation still relies primarily on a fallback analytic simulator rather than on a full physical stack or external empirical validation dataset. Although a small-batch physical probe is now available, its alignment is currently strongest for EUIt, weaker for the simplified EG proxy, and still more limited for H than for EUIt because the current H path is a small-batch windowsill-style direct-sun-hours reconstruction rather than a full publication-pipeline recreation. In particular, the current physical H evidence should be read as a winter-specific direct-sun-hours check over the January 20 design window, not as a validated annual daylight or comfort metric. Fourth, the reevaluation shows that both DDPG and NSGA-II candidates can drift materially relative to their surrogate predictions, so surrogate-space ranking should not be overgeneralized. Fifth, the newly added checkpoint-sensitivity audit indicates that even two surrogates with similar training-fit metrics can differ in extrapolative bound-violation behavior, and the broader three-context summary shows that benchmark ordering itself can change across checkpoint selections; even so, the current evidence still covers only a small number of checkpoints rather than a full multi-checkpoint ranking study. Sixth, although a representative local CPU-only wall-clock and memory audit is now available, it is still a single-machine reference rather than a hardware-normalized production benchmark across heterogeneous compute environments. Seventh, although the current comparison now includes a CMA-ES follow-up, entropy-regularized actor-critic baselines such as SAC remain untested here; therefore, the current policy-learning conclusions should be read as applying most directly to the tested deterministic policy-gradient setup rather than to all actor-critic methods. Eighth, the present RL formulation is a scalarized static black-box optimization problem recast as policy learning; therefore, alternative continuous optimizers remain important baselines for future work. Ninth, the strong physical performance of the RandomSearch representative indicates that some of the physically projected design region may be relatively flat or may be compressed by the current nearest-block probe. Accordingly, RandomSearch is used here as an uninformed physical reference rather than as a fully developed optimizer benchmark with the same analytical depth as DDPG or NSGA-II. Tenth, the four-method physical probe in Appendix B is still a representative-candidate comparison rather than a fully symmetric benchmark: DDPG, CMA-ES, and RandomSearch are shown through balanced-scenario representatives, whereas NSGA-II is shown through the selected benchmark representative. It therefore strengthens the validation-mode argument, but it does not replace the main surrogate-space benchmark reported in Section 3.

5. Conclusions

This study presents a DRL-DNN framework for co-optimizing urban morphology and energy performance in a surrogate-based setting. The evidence suggests that the resulting paper should be interpreted primarily as a benchmark-fragility study in surrogate-assisted urban energy design rather than as a demonstration of policy-learning superiority.

Three main conclusions can be drawn. First, the dataset-scale study identifies the 2000-sample DNN with standard scaling as the most accurate surrogate among the tested candidates. Second, when the optimizer comparison is conducted on that surrogate within the paper's three-scenario DDPG-versus-NSGA-II architecture, fair-budget NSGA-II outperforms all three DDPG scenarios in archive-level objective summaries, Hypervolume, IGD, and preference-aligned utility. Third, the DDPG scenarios still yield interpretable morphology fingerprints, but those differences should be treated as surrogate-conditional diagnostics rather than as evidence of a robust policy-learning advantage.

Overall, this paper shows that optimizer ranking in surrogate-assisted urban energy design can remain fragile even after moving to a higher-accuracy surrogate and a fair-budget comparison. The resulting workflow is best viewed as a diagnostic methodology for bounding benchmark claims by surrogate reliability, benchmark fairness, and validation mode rather than as a leaderboard of optimizer quality. The supplementary checkpoint audit, CMA-ES follow-up, and representative physical probe do not overturn the main benchmark result; instead, they show why that result must be interpreted conditionally rather than promoted as a universal optimizer ranking. In practical terms, the paper now supports a four-step screening rule for future studies: audit out-of-bounds rates, compare more than one checkpoint, reevaluate apparent corner optima independently, and treat deterministic convergence to a clipped corner as a surrogate-risk signal rather than as an immediate design recommendation. Future work should expand the checkpoint-sensitivity analysis, test entropy-regularized RL baselines such as SAC explicitly rather than only citing them, and validate the final solutions through direct simulation and external data.

Appendix A. Additional Case Settings and Surrogate Validation

Appendix A.1. Case-setting details

The base urban block is represented as a 240 m × 240 m square subdivided into a 3 × 3 grid of nine land units. Each land unit measures 80 m × 80 m. A two-step inward offset is applied to represent the road boundary and building setback, resulting in a nominal buildable area of 60 m × 60 m for each unit. One land unit is designated as open space, and the remaining units are populated with residential building prototypes.

To avoid notation drift between the main text and the appendix, the following definitions are used consistently:

- **AF** (average floors) is defined as the total gross floor area divided by the total building footprint area.
- **OSR** (open space ratio) is defined as the non-built-up area of the block divided by the total gross floor area.

Table A.8: Representative building prototypes used in the parametric dataset.

ID	Type	Footprint Area (m^2)	Floors	FAR Range
P-1	Point	2000	1–3	0.31–0.94
P-2	Point	1500	4–12	0.94–2.81
P-3	Point	1000	13–30	2.03–4.69
P-4	Point	1000	13–30	2.03–4.69
S-1	Slab	2000	1–3	0.31–0.94
S-2	Slab	1500	4–12	0.94–2.81
S-3	Slab	1000	13–30	2.03–4.69
C-1	Courtyard	2000	1–3	0.31–0.94
C-2	Courtyard	1500	4–12	0.94–2.81

Appendix A.2. Surrogate validation

The original surrogate workflow began from a 500-sample baseline, but the final optimizer comparison in this revision uses the strict highest-accuracy surrogate obtained from the expanded 2000-sample regime. In the synced artifact bundle, that regime is generated under the documented `fallback_analytic` simulation mode rather than by a direct full-stack EnergyPlus/Radiance rerun. Five-fold cross-validation is used to assess that final selected model. Table A.9 reports the corresponding aggregate validation results.

Table A.9: Five-fold cross-validation summary for the DNN surrogate model.

Objective	Mean R^2	Mean MAE	Std. R^2	Std. MAE
EUIt	0.983	0.209	±0.001	±0.005
EG	0.972	0.026	±0.002	±0.001
H	0.992	0.025	±0.001	±0.001

To evaluate reliability in optimization-relevant regions, additional error slices were computed over the lower and upper tails of each target distribution. The corresponding mean absolute errors are listed in Table A.10. Table A.11 reports complementary surrogate-fidelity statistics from the cross-validation predictions.

Table A.10: Error diagnostics in extreme regions of the target distributions.

Target	$\mathrm{MAE}_{\mathrm{all}}$	MAE_{q10}	MAE_{q90}	$q_{0.10}$	$q_{0.90}$
EUIt	0.209	0.283	0.194	67.83	73.32
EG	0.026	0.027	0.030	2.132	2.655
H	0.025	0.026	0.029	6.514	7.429

Table A.11: Additional surrogate-fidelity audit on the cross-validation predictions of the selected 2000-sample surrogate.

Target	RMSE	MAE	nRMSE	Tail nMAE	R^2
EUIt	0.279	0.209	0.0227	0.0194	0.983
EG	0.033	0.026	0.0305	0.0265	0.972
H	0.032	0.025	0.0172	0.0150	0.992

Note: nRMSE is normalized by the observed range of each target over the cross-validation predictions. Tail nMAE is the mean of the lower-tail and upper-tail nMAE diagnostics used for surrogate selection.

Appendix B. Benchmark Robustness and Independent Re-evaluation

Appendix B.1. Training dynamics and benchmark protocol

The revised server-side reruns used for the final accuracy-first comparison were executed under the shared guarded surrogate evaluator to suppress exploitation of extrapolative edge cases. On the strict highest-accuracy 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401 (balanced), 320 (saving), and 398 (generation). Compared with the earlier lower-scale rerun, the balanced scenario becomes visibly smoother and less noisy across seeds, but the saving-focused and generation-focused scenarios do not show a uniform reduction in cross-seed variability across reward and objective trajectories. However, late-stage regression remains common: the mean best-to-final reward gap ratios are about 0.62, 0.71, and 0.71 for the balanced, saving, and generation scenarios, respectively, and the substantial-regression fractions remain high at 95%, 85%, and 100%.

For benchmark comparison, the revised implementation removes the previous benchmark-matching shortcut from the publication-mode NSGA-II path. Publication mode compares DDPG and NSGA-II under a shared surrogate environment, matched per-seed evaluation budget, and matched seed count, and applies a post-hoc preference-aligned utility summary consistently across methods. On the strict highest-accuracy 2000-sample surrogate, NSGA-II remains stronger than all three DDPG scenarios. The corresponding budget accounting and objective-space separation are summarized in Table B.12.

Table B.12: Budget accounting and objective-space separation in the highest-accuracy 2000-sample rerun.

Item	Value 1	Value 2	Interpretation
DDPG per-seed budget	600 episodes	40 steps/episode	24,000 surrogate interactions
DDPG seeds per scenario	20	–	pooled into one scenario archive
NSGA-II seeds in rerun	20	–	pooled into one comparison archive
Distance: balanced vs generation	0.5417	–	normalized objective-space distance of scenario means
Distance: balanced vs saving	0.6809	–	normalized objective-space distance of scenario means
Distance: generation vs saving	0.7619	–	normalized objective-space distance of scenario means
NSGA-II archive spread	0.1642	0.1968	mean and std of distance to normalized archive center

Note: The 2000-sample highest-accuracy rerun matches both the per-seed interaction budget and the number of seeds between DDPG and NSGA-II. The DDPG scenarios are more separated than in the earlier matched-checkpoint run, but NSGA-II still dominates the preference-aligned utility comparison.

To probe whether this fragility is visible even before comparing optimizer archives, an additional checkpoint-sensitivity audit was run on two available 2000-sample surrogate checkpoints: the imported publication checkpoint and a locally retrained checkpoint synced from the same revision branch. Table B.13 shows that the two checkpoints have very similar training-fit statistics, yet they do not behave identically when evaluated on random samples drawn from the observed feature ranges. In particular, the locally retrained checkpoint produces a larger fraction of EUIt predictions below the observed minimum (0.0080 versus 0.0008) and a larger fraction of H predictions above the observed maximum (0.0040 versus 0.0012). The fraction of random samples that simultaneously exceed the observed target bounds also changes from 0.0004 to 0.0006. These differences are small in absolute terms, but they support the narrower diagnostic claim that similar in-sample fit does not guarantee identical extrapolative behavior.

Table B.13: Checkpoint-sensitivity audit for two 2000-sample surrogate checkpoints.

Metric	Imported checkpoint	Local retrained checkpoint	Interpretation
EUIt training MAE	0.2038	0.1885	similar training fit
EG training MAE	0.0248	0.0245	similar training fit
H training MAE	0.0252	0.0261	similar training fit
EUIt below observed min fraction	0.0008	0.0080	local checkpoint shows higher low-side extrapolation risk
EUIt above observed max fraction	0.0646	0.0674	both checkpoints extrapolate above the observed upper bound
H above observed max fraction	0.0012	0.0040	local checkpoint shows higher high-side extrapolation risk
Triple better-than-bounds fraction	0.0004	0.0006	both checkpoints admit rare optimizer-exploitable regions

Note: The audit compares the imported publication checkpoint and one synced local retrained checkpoint on the 2000-sample dataset using 5000 random feature samples drawn from the observed feature ranges.

Table B.14 complements that audit at the benchmark level. It compares the imported publication artifact with two later checkpoint selections that were retained during the revision process. The imported publication artifact still favors DDPG in Hypervolume, whereas the strict-highest-accuracy 2000-sample checkpoint and the surrogate-rebenchmark checkpoint both favor NSGA-II in both Hypervolume and IGD. The purpose of this table is not to promote one checkpoint as universally correct, but to show that optimizer ordering itself changes materially with surrogate selection.

Table B.14: Benchmark-ordering fragility across three available checkpoint contexts.

Checkpoint context	Best DDPG HV	NSGA-II HV	Best DDPG IGD	NSGA-II IGD	Ordering takeaway
Imported publication artifact	1.2696	0.6528	0.5043	0.1419	HV favors DDPG, but IGD already favors NSGA-II
Strict-highest-accuracy 2000 checkpoint	1.0107	1.3310	0.2205	0.1068	NSGA-II favors in both HV and IGD
Surrogate-rebenchmark checkpoint	0.9337	1.3310	0.2874	0.1036	NSGA-II favors in both HV and IGD

Note: "Best DDPG" denotes the strongest of the three DDPG scenarios within each checkpoint context. The first row comes from the imported publication artifact, while the other two rows come from the synced strict-highest-accuracy and surrogate-rebenchmark checkpoint bundles.

Finally, to test whether this surrogate fragility is specific to DDPG or can also affect a stronger continuous optimizer, a same-budget CMA-ES follow-up was run locally on the three checkpoint contexts summarized above. Table B.15 reports the result. Across all three contexts, all three scalarization scenarios, and all ten seeds per scenario, CMA-ES converges to the same clipped corner solution (66.0, 2.85, 7.85) with reward 1.0. A trajectory probe further shows that this is not an immediate one-step failure: in a representative balanced-scenario run, the first perfect-reward hit occurs only after about 2992–3168 evaluations across the three checkpoint contexts, after a gradual rise from initial rewards around 0.35–0.44. By contrast, the same-budget random-search archive available for the imported publication artifact remains materially below this corner, with mean rewards around 0.87 rather than 1.0. Deterministic analytic re-evaluation of representative CMA-ES corner candidates also shows heterogeneous drift across checkpoints: the imported publication and strict-highest-accuracy checkpoints retain EUIt = 66.0 but lose between about 0.06 and 0.21 on EG/H, whereas the surrogate-rebenchmark checkpoint degrades more strongly to EUIt = 68.66, EG = 2.56, and H = 7.45. This result does not establish CMA-ES as a better optimizer in the present surrogate workflow; rather, it indicates that once the surrogate exposes a reward-saturating corner, a strong continuous optimizer can exploit the same artifact deterministically, while independent analytic re-evaluation still reveals checkpoint-dependent overstatement at the corner. The methodological implication is that the central risk is broader than DDPG instability alone.

Table B.15: Same-budget CMA-ES follow-up across three checkpoint contexts.

Checkpoint context	EUIt	EG (10^6 kWh/y)	H (h)	Reward	Observation
Imported publication artifact	66.0	2.85	7.85	1.0	identical optimum across all three scenarios and all 10 seeds per scenario
Strict-highest-accuracy 2000 checkpoint	66.0	2.85	7.85	1.0	identical optimum across all three scenarios and all 10 seeds per scenario
Surrogate-rebenchmark checkpoint	66.0	2.85	7.85	1.0	identical optimum across all three scenarios and all 10 seeds per scenario

Note: Each row summarizes a same-budget CMA-ES run with 24,000 surrogate evaluations, 10 seeds per scenario, and three reward-scalarization scenarios. The reported objective tuple is the scenario-level optimum reached in every run.

To complement the linear correlation results in the main text, Fig. B.12 reports representative surrogate-based nonlinear response profiles for OSR→EUIt, FAR→EG, SVF→H, and θ →H.

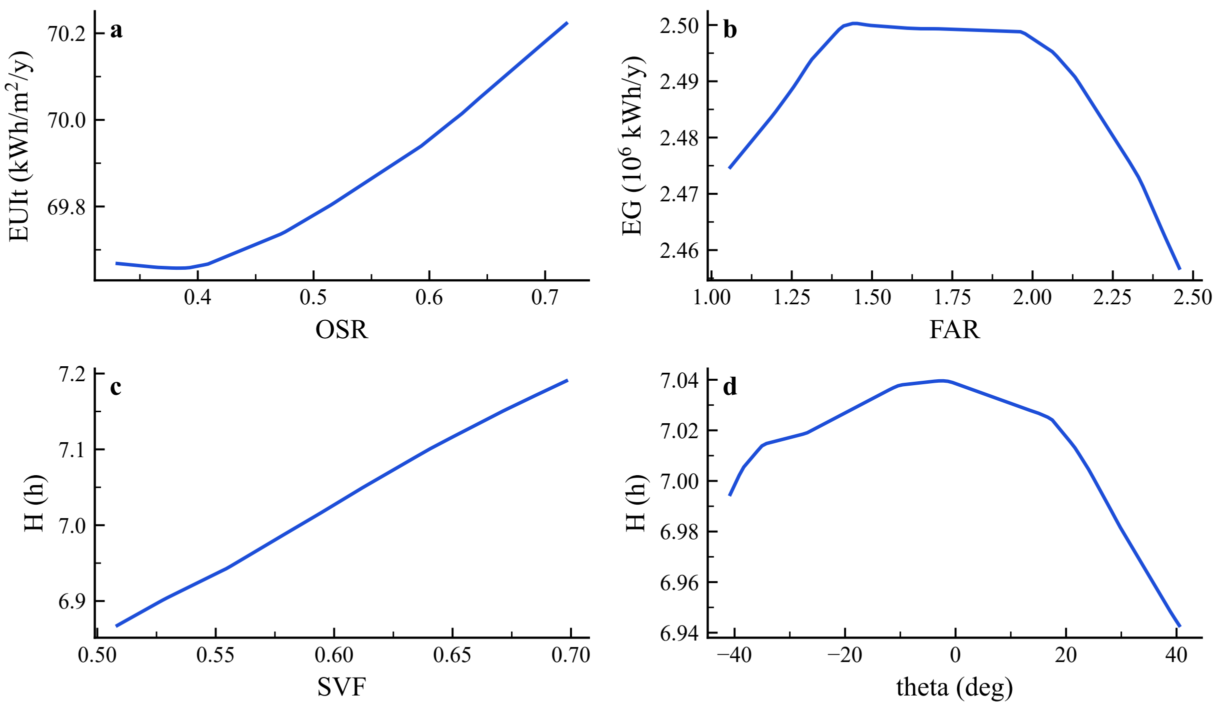

Figure B.12: Surrogate-based nonlinear response profiles for selected UMF-target pairs: (a) OSR→EUIt, (b) FAR→EG, (c) SVF→H, and (d) θ →H.

Appendix B.2. Independent re-evaluation of selected candidates

To reduce over-reliance on surrogate predictions, the workflow re-evaluates preference-aligned candidate solutions using an independent simulation call. In the current local environment, this re-evaluation still uses the fallback analytic simulator, but the code path is structured so that a full physical-stack simulation can replace it when server-side results become available. Table B.16 reports the latest available reevaluation results for the strict highest-accuracy 2000-sample comparison. The updated evidence shows heterogeneous drift: the balanced DDPG winner improves in EUIt but loses EG and H, the saving-focused and generation-focused DDPG winners drift upward in EUIt and downward in EG, and the selected NSGA-II candidate also moves away from its surrogate prediction. This narrows the earlier asymmetry between DDPG and NSGA-II, but it still indicates that surrogate-space ranking should not be treated as a direct substitute for independent validation.

Table B.16: Independent re-evaluation of selected candidates from the highest-accuracy 2000-sample surrogate rerun. Surrogate and deterministic re-evaluation values are reported for the current preference-aligned selections.

Method	Scenario	$EUIt_{sur}$	$EUIt_{re}$	EG_{sur}	EG_{re}	H_{sur}	H_{re}
DDPG	Balanced	67.168	66.000	2.751	2.689	7.697	7.374
NSGA-II	NSGA-II (seed 20260310)	66.000	67.730	2.850	2.712	7.850	7.510
DDPG	Saving	66.698	67.473	2.844	2.769	7.671	7.611
DDPG	Generation	66.356	67.836	2.841	2.735	7.807	7.530

To complement the deterministic fallback reevaluation, a small-batch physical-stack probe was also carried out on two representative candidates using the repo-side physical-stack path. The current best-aligned version of that path combines a compressed thermal model for EUIt, a simplified rooftop-PV proxy for EG, and a January 20 windowsill-style direct-sun-hours check for H. Table B.17 summarizes the result. The main value of this table is not to claim full physical closure, but to show that the physical probe has now moved beyond proof-of-life and into a partially aligned diagnostic regime. In particular, the two representative candidates yield physical EUIt values of 74.379 and 74.057 kWh/m^2/y, which are clearly above but still close to the corresponding surrogate values of 66.000 and 69.443. The EG proxy also remains directionally close to the surrogate values, and the revised H check is now in the same overall range as the surrogate-scale sunlight-hours values, although it remains less stable across candidates than EUIt. The methodological implication is therefore still asymmetric: the present physical probe provides the strongest check for EUIt, a weaker but stable structural check for EG, and a newly improved but still candidate-sensitive direct-sun-hours check for H.

Table B.17: Small-batch physical-stack probe for two representative candidates under the current best-aligned settings.

Method	Scenario	$EUIt_{sur}$	$EUIt_{phys}$	EG_{sur}	EG_{phys}	H_{sur}	H_{phys}
DDPG	Balanced	66.000	74.379	2.850	1.760	7.850	7.525
NSGA-II	NSGA-II	69.443	74.057	2.613	2.479	7.569	6.075

Note: ‘phys‘ denotes the current project-level `physical_stack_probe`, not a final publication-grade reconstruction of all three paper metrics. EUIt is the strongest-aligned physical quantity under the current settings. EG is derived from a simplified rooftop-PV proxy. H is now derived from a windowsill-style January 20 direct-sun-hours calculation, which is much closer to the manuscript definition than the earlier point-illuminance proxy but should still be interpreted as a small-batch partial closure rather than as a full publication-pipeline replacement.

The same best-aligned physical settings were then extended to a four-method small-batch probe spanning the representative DDPG, NSGA-II, CMA-ES, and RandomSearch candidates that had previously been used for debugging the physical stack. For DDPG, CMA-ES, and RandomSearch, the representative points come from the balanced scalarization setting, whereas NSGA-II is represented by the selected benchmark candidate used elsewhere in the revision package. Here, RandomSearch is included not as a co-equal optimizer benchmark with the same analytical depth as DDPG or NSGA-II, but as an uninformed physical reference for how much spread remains after projection through the current probe. Under this broader check, all four methods remained executable through the same project-level path and the resulting physical EUIt values collapsed into a relatively narrow band of about 73.3–74.4 kWh/m^2/y. The corresponding windowsill-style direct-sun-hours values also narrowed to about 6.0–7.5 h. Figure B.13 visualizes the same point by directly pairing surrogate and physical-probe values for the four representative candidates. This broader probe does not replace the surrogate benchmark reported in the main text, but it strengthens the narrower methodological interpretation: once the comparison is projected through the current physical probe, the method-level spread becomes smaller than the surrogate-space spread. The current evidence does not yet determine whether this shrinkage is caused mainly by a genuinely flat physical design region, by the nearest-block restriction of the probe, or by the surrogate's tendency to exaggerate method-level differences. It is therefore safest to interpret the result as evidence that optimizer ranking remains conditional on surrogate choice and validation mode, not as proof that all methods are physically equivalent.

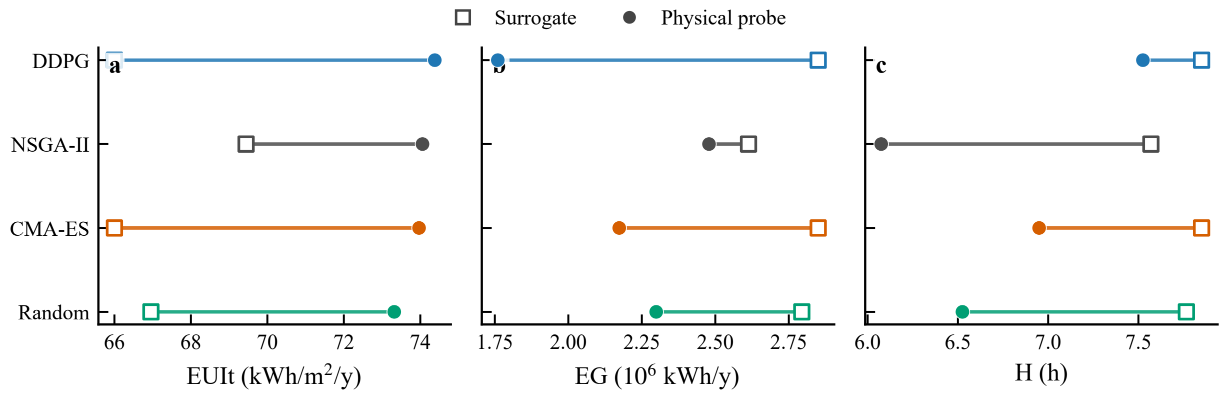

Figure B.13: Four-method small-batch comparison between surrogate predictions and the current best-aligned physical probe. Each row shows one representative candidate for DDPG, NSGA-II, CMA-ES, and RandomSearch, with square markers for surrogate values and circular markers for physical-probe values. The DDPG, CMA-ES, and RandomSearch rows use balanced-scenario representatives, while the NSGA-II row uses the selected benchmark representative. The physical projection compresses the method-level spread most clearly for EUIt and also narrows the H range relative to the surrogate values, while EG remains directionally informative but should still be read as a simplified rooftop-PV proxy.

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